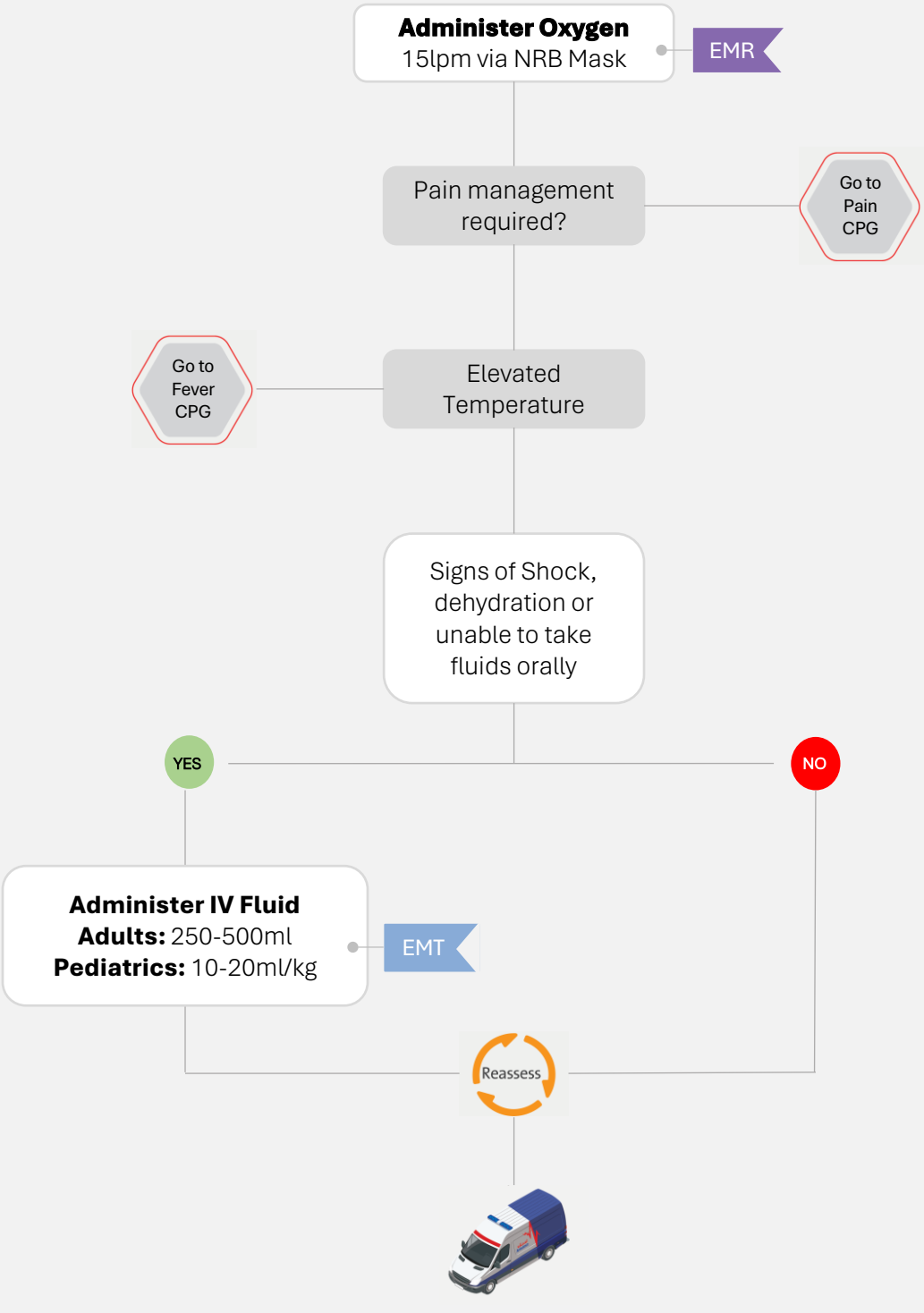

Patient Safety Considerations

- Do not delay pain relief — early treatment improves patient outcomes.
- Avoid aggressive fluid administration in patients with chest pain symptoms.
- Monitor closely for deterioration (e.g., increasing respiratory rate, reduced GCS).
- Use weight-based dosing; refer to Broselow tape for pediatric patients.
- Avoid causing stress or cold exposure, which may precipitate crisis.

Patient Care Goals

1. Rapidly identify and assess patients presenting with a sickle cell crisis
2. Manage pain effectively and prevent further sickling events
3. Maintain adequate oxygenation, hydration, and perfusion
4. Identify complications requiring urgent hospital intervention (e.g., acute chest syndrome, stroke)
5. Ensure safe and timely transport with appropriate prehospital interventions

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Known diagnosis of Sickle Cell Disease (SCD)
- Presenting with symptoms suggestive of acute vaso-occlusive crisis (VOC), including:
  - Sudden, severe bone or joint pain
  - Chest pain or shortness of breath
  - Abdominal pain
  - Headache or neurological symptoms (suggestive of stroke)
  - Fever

Exclusion Criteria

- Pain from unrelated causes (e.g., trauma) without SCD history
- Known bleeding disorder unrelated to SCD
- No signs of crisis or acute complications

Key Patient Assessment Points

1. **General Impression**
   - Use the Pediatric or Adult Assessment Triangle for general impression (depending on age).
   - Look for signs of distress, pallor, respiratory difficulty, or altered mental status.
2. **Airway & Breathing**
   - Assess for tachypnea, hypoxia, or signs of acute chest syndrome.
   - Auscultate for decreased breath sounds or crackles.
3. **Circulation**
   - Check capillary refill, heart rate, blood pressure, and perfusion.
   - Look for signs of dehydration or hypotension.
   - Assess for pallor, cyanosis, or delayed refill.
4. **Pain Assessment**
   - Assess for pain as per Pain Management CPG.
5. **Neurological Assessment**
   - Evaluate for headache, confusion, focal deficits, or seizure activity (possible stroke).
6. **History (SAMPLE/OPQRST)**
   - Past similar episodes, baseline functional level
   - Medications (e.g., hydroxyurea, analgesics)
   - Last crisis or transfusion
   - Any recent illness or dehydration
   - Fever or infection risk
7. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
   - Request ALS if patient has altered mental status, respiratory distress, signs of stroke, or requires IV opioid management.
2. **Pain Management**
   - Reassure and position for comfort
   - Administer appropriate analgesia early as per Pain Management CPG.
     - IV analgesics are likely to be required (e.g., IV Morphine, IV paracetamol)
   - Avoid delays in pain relief — consider escalation to ALS for opioid administration.
3 **Oxygen Therapy**
   - Administer oxygen if SpO_2 <94% or signs of respiratory compromise
   - Target SpO_2 ≥94%; avoid hyperoxia
   - Monitor continuously with pulse oximetry
4. **IV Access & Fluids**
   - Establish IV access
5. **Fluid Resuscitation**
   - Indications:
     - Signs of dehydration, fever, vomiting, or hypotension,
   - Administer NaCl:
     - Adults: 250ml IV bolus, may be repeated if required.
     - Pediatrics: 10-20mL/kg, IV bolus, may be repeated if required.
   - Avoid large-volume fluid boluses unless the patient is hypotensive.
   - If acute chest syndrome is suspected, fluid administration should be minimized and closely monitored.
   - Titrate fluids to improve perfusion rather than normalize blood pressure.
6. **Monitor for Complications**
   - Consider alternative causes of pain and symptoms (e.g. ACS, Stroke, Sepsis). Provide additional treatment as per relevant CPG.
7. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.

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Clinical Insights & Additional Guidance

- Sickle cell crises are painful and time-sensitive emergencies; early pain relief is critical
- Many patients have individualized treatment plans; involve caregivers when appropriate
- Recurrent pain does not mean drug-seeking; treat all reported pain seriously
- Acute chest syndrome is the most life-threatening complication; recognize and report early
- Fever in a sickle cell patient is always a red flag for possible sepsis
- If in doubt, consult Tele-EMS or request ALS backup for escalation

Documentation

- Identification: Name, age, sex, known diagnosis of SCD
- Medical complaint: Pain location, onset, duration, and severity, chest pain, fever, or neurological symptoms
- Information: Onset of crisis, prior episodes, current medications

- Signs: Vital signs, SpO_2, pain score, LOC
- Treatment: Oxygen, analgesia, IV fluids, ALS request
- Allergies: Document if known
- Medications: Prehospital medications and those used by patient
- Background: Any history of previous similar episodes, home analgesia, or hospital admissions for pain, transfusions, prior complications
- Other: Response to treatment, ETA to hospital, family/caregiver concerns.

Key Performance Indicators

1. Assess and document pain using an appropriate pain scale.
2. Administer appropriate analgesia within 15 minutes of arrival.

References

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